A binder for the KRAS G12D neoepitope on HLA-A*11:01: what twenty designs rest on, and what they do not

Anthropic × Adaptyv Protein Design Competition, Challenge 5

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Abstract

The challenge asks for a protein that binds the KRAS G12D peptide `VVVGADGVGK` presented on HLA-A*11:01 and not the wild-type `VVVGAGGVGK` on the same allele, ranked on mutation specificity first. We report that the first tier is not reachable with the tools available, and we report it as a measurement rather than as a difficulty. The one atom that chemically distinguishes the two targets, the G12D carboxylate oxygen OD1, has a **0.013%** escape cone at 8 Å reach — 34 of 262,144 rays — so no designed side chain can touch it. The discrimination that remains is indirect: a 0.85/0.73/0.48 Å backbone shift at peptide positions 6/7/8, against a 0.337 Å superposition residual, which is below the coordinate error of the co-fold predictors used. Measured on the one receptor known to discriminate these peptides, the predictor's wild-type-versus-mutant sign is not merely unresolved but *inverted*, by $+0.0027 \pm 0.0006$ over eight independent invocations.

We therefore designed for the geometry that is reachable — the conformational bulge at peptide positions 5–8 — and ranked on peptide-*dependence*: how much a predicted interface degrades when the peptide is removed from the groove. Calibrated against the 7OW6 receptor over ten independent invocations the bar is +0.106. Of the 78 screened designs, measured twice independently at five diffusion samples per condition, **0 are above that bar, 4 are tied with it, and 74 are below**; 14 of 78 clear zero. Twenty designs are submitted, which is the cap. We state the funnel with every denominator, declare which filters were fixed after seeing results, and list what is not established — including that only one co-fold predictor was used, so the project's own multi-method requirement is unmet.

1 The target, measured before anything was designed

The inputs were frozen from three depositions of the same system: 7OW6 (G12D ternary complex, HLA heavy chain A, β_2m B, peptide C, TCR D/E), 7OW5 (wild type), and 7OW4 (receptor-free). The TCR chains were deleted for the generation target. One heavy chain and one β_2m serve all three co-fold conditions, which differ only in the peptide: the two depositions disagree by a leading construct residue (276 against 275), and carrying that difference into the comparison would have placed a sequence change inside the exact contrast selectivity is read from.

Two of these are load-bearing. The 161-fold collapse of the escape cone between 6 and 7 Å locates the carboxylate at the bottom of a pocket about 6 Å deep; the probe controls behaved as a working measurement should, with an exposed neighbouring carbonyl at 39.1% and a buried one at 0.00%. And the 0.85 Å backbone shift sits only 2.5 times above the residual of the superposition used to measure it, so a predictor whose coordinate error is of that order cannot be expected to resolve it.

An earlier version of the escape-cone number was quoted as 0.02% of 4096 rays. That is a single surviving ray — the sampling floor, not a measurement — and it was the most pessimistic corner of a probe × reach grid that had not been run. The grid has since been run: the conclusion did not change, but the evidence for it did, and the tool now reports the surviving ray count beside the fraction so a result resting on one ray cannot be quoted again without that being visible.

Table 1: Decisive target measurements, all taken before any design was generated.

Quantity	Condition	Value
Peptide share of complex accessible surface	—	1.2% (239.5 of 19,612 Å^2)
Position-6 side-chain SASA (mutant)	—	8.8 Å^2 (OD1 6.43, OD2 0.00)
Escape cone over OD1	8 Å reach, 1.55 Å probe	**0.013%** (34 / 262,144 rays)
Escape cone over OD1	3 / 4 / 5 / 6 / 7 Å	41.0 / 21.9 / 7.1 / 2.9 / 0.018%
Escape cone, position-7 carbonyl	8 Å (probe control)	39.1%
Escape cone, position-4 carbonyl	8 Å (probe control)	0.00%
TCR contacts to position 6	5.5 Å	17, *all* to backbone
TCR interface split	4.5 Å	55 peptide / 220 HLA (share 0.200)
Peptide backbone shift, mutant vs. wild type	Cα, positions 6/7/8	0.85 / 0.73 / 0.48 Å
HLA superposition residual	the baseline that shift sits against	0.337 Å

2 What the first ranking tier would require, and why it fails

Mutation specificity is the competition's first tier. The strongest available test of whether any co-fold predictor can supply it is to run the test on a molecule whose answer is known: the TCR in 7OW6, which was crystallised bound to the G12D complex and is specific by construction. If the predictor scores that receptor higher against the mutant peptide than against the wild type, a design's score in the same column means something. If not, it does not.

Paired within each of the eight independent invocations that predicted the wild-type condition, the receptor scores **wild type above G12D by** $+0.0027 \pm 0.0006$, positive in seven of the eight. The sign is wrong. This is a stronger statement than "unresolved": the scoring function is anti-correlated with the quantity tier 1 asks for, on the one molecule that can calibrate it. Every subsequent use of the wild-type column in this work is therefore reported and never gated.

3 Method

Generation used BoltzGen 0.1.4 against the intact complex, in four hotspot variants of 64 designs each — `bulge_core` (peptide positions 5,6,7), `bulge_wide` (4–8), `mut_proximal` (5,6) and `hub` (7,8). The denominator stays **256** in every funnel below, including designs rejected for not being designs.

Verification is Boltz 2.2.1, independent of the generator's own folding, in three conditions per design: `complex_g12d` (binder + HLA + β_2m + mutant peptide), `apo_hla` (the same without the peptide) and `complex_wt`. The ranked quantity is

$$\Delta\text{ipTM} \;=\; \text{ipTM}(\texttt{complex_g12d}) - \text{ipTM}(\texttt{apo_hla}), \tag{1}$$

the degradation of the predicted interface when the peptide leaves the groove. It is a proxy for peptide-dependent binding, the competition's second tier, and it is explicitly not affinity.

3.1 The filters, and when each was fixed

The engagement threshold is the one that is genuinely calibrated rather than chosen: 0.200 is the peptide's share of the 7OW6 receptor's own interface, measured with the same code on the same cutoff. A design that puts less area on the peptide than the natural receptor does cannot be reading the peptide.

Table 2: Gates in order of application. "After" means the criterion was set after results were seen; no statistic derived from such a gate is presented as pre-registered.

Gate	Criterion	Fixed
novelty	$< 30\%$ of the design is a verbatim copy of a target or TCR chain, by local alignment	after
fold novelty	top FoldSeek hit is neither a known protein at high identity nor the target's own immunoglobulin β-sandwich class	after
engagement	peptide share ≥ 0.200 and every nominated hotspot touched	criterion before; threshold calibrated on the 7OW6 receptor
peptide-dependence	ΔipTM measured at five samples per condition	before
selectivity	`complex_wt`	**reported, not gated**
ranking denominator	the cohort's pooled between-run SD, not a design's own two runs	after

3.2 The novelty incident, and a claim withdrawn

A two-design pilot against a *truncated* target returned β_2-microglobulin — a chain of the target itself — at 90.5% and 85.7% identity. Truncation had left β_2m's own interface vacant and the generator filled it with the protein that belongs there. Generation moved to the intact complex, which reduced the rate from 2 of 2 to 17 of 256 but did not eliminate it, so novelty is a hard gate applied before selection rather than a precaution.

The gate itself was wrong when first shipped. It used a global alignment normalised by the shorter sequence, which cannot represent a truncated copy at all: a verbatim 40-residue slice of the 276-residue heavy chain scored **0.000**, and a 60-residue slice of the TCR β chain, the receptor the rules forbid modifying, also scored 0.000. Both would have passed as novel. The replacement is the fraction of the design that is a verbatim copy of some target chain by local alignment, normalised by the design's own length; on the same controls it scores 1.000. It was chosen on synthetic controls alone, independent of the cohort, and the superseded value is kept in the output so the change is auditable.

One justification offered for gating novelty first was that ipTM rewards plagiarism threefold, so a score-first ranking would select inadmissible designs preferentially. Measured on the pilot's four designs that was true; measured over all 256 it is not. The ratio of medians is **0.83**$\times$ — the β_2m copies score slightly *worse* than admissible designs, 39 of 188 admissible designs beat the best plagiarised one. The threefold effect was real but local to target preparation. The claim is withdrawn; novelty still gates first, on correctness rather than on bias.

4 The funnel

All twenty slots are used. Four designs are tied with the bar and none is above it, so sixteen of the twenty are submitted as the best available rather than as designs that met a criterion. That is stated here rather than left to be inferred from a ranked list.

5 The measurement of record

The first ranking used one diffusion sample per condition. A 26-design probe at five samples *reversed the sign* of two of its top four, which is what a difference of two single draws should be

Table 3: Every denominator. The campaign's result is this table, not its first row.

Stage	Surviving	Removed by
generated (4 variants × 64)	256	—
sequence novelty	239	17 verbatim β_2m copies, 30.3–92.6% copied
fold novelty	188	16 copies of other known proteins (TM 0.93–0.98), 35 immunoglobulin folds
engagement	78	110 below the receptor's 0.200 peptide share, or missing a hotspot
co-folded twice, 5 samples	78	— (318 of 318 predictions scored, 0 failures)
clears zero on ΔipTM	14	64 whose dependence is indistinguishable from none
at or above the receptor's bar	**4**	10 below the bar but above zero
submitted	**20**	cap is 20; slots 5–20 are the best of the remainder

expected to do, so that screen is retrospectively a pilot. Everything below is five samples per condition, and the whole 159-spec cohort pass was run twice independently.

5.1 The bar, over ten independent invocations

Table 4: The 7OW6 receptor's ΔipTM, measured ten times from byte-identical specifications. Boltz seeds once per process, so samples within one invocation share that state; separate invocations are the only way to observe between-run variation.

Measurement	ΔipTM
8 small-batch invocations	+0.101, spread 0.004 (span +0.095 to +0.108)
26-spec probe	+0.104
159-spec cohort, run 1	**+0.150**
159-spec cohort, run 2	+0.101
pooled over all ten	**+0.106**, spread 0.016
nine, excluding run 1's control	+0.101

Nine of ten measurements of the same molecule agree and one does not. That run is excluded from the +0.101 figure on a *structural* criterion measured independently of its score, not on the score itself: its peptide-free prediction scored pTM 0.836 against 0.877 and receptor-to-HLA pair ipTM 0.710 against 0.786, while pLDDT barely moved and the complex condition reproduced to three decimal places. It is a docking failure in one condition of one run.

Pairing each design against its own run's control would have been the obvious alternative to excluding it, and it was tested rather than assumed. Across the twelve designs present in both runs, design apo scores moved $+0.0006 \pm 0.0150$ while the control's fell 0.045. The shift was not shared, so pairing would have charged every design for one bad prediction.

5.2 Two variance components, and the one that is not there

A set of run means has a variance that *already contains* the within-run sampling error, so the two subtract rather than add:

$$\sigma_{\text{run}} \;=\; \sqrt{\max\!\left(0,\; \sigma^2_{\text{between}} - \text{SEM}^2_{\text{within}}\right)}. \tag{2}$$

On the control, $\sigma_{\text{between}} = 0.004$ against $\text{SEM}_{\text{within}} = 0.007$ gives a negative variance, and a 20,000-draw permutation test over the pooled samples puts the observed spread at the **4th**

percentile of the no-run-effect null. There is no resolved between-run component for the control, and its single-run standard error is if anything conservative. An earlier version of this analysis combined the two in quadrature; that double-counted and is withdrawn.

Designs behave differently: pooled between-run SD **0.018** against a within-run error of 0.015, leaving a run effect of 0.009. Every per-design verdict divides by the pooled figure over $\sqrt{\text{runs}}$, because two runs cannot estimate one design's own spread — the estimate carries roughly 70% relative error, and a pair that happens to agree reports 0.000, which would make a +0.011 drop read as significant and a reproducible +0.112 read as infinitely so. The control is excluded from the pool it is the bar for, so the bar cannot widen its own margin.

5.3 The result

Table 5: Peptide-dependence, pooled over two independent invocations.

	ΔipTM	per-run values
`mut_proximal_16`	+**0.112**	+0.112, +0.112 (between-run SD 0.0001)
`mut_proximal_26`	+0.095	+0.109, +0.081
the 7OW6 receptor, ten runs	+0.106	—
median of 78	−0.010	—
worst of 78	−0.093	—

0 designs above the bar, 4 tied, 74 below. 14 of 78 clear zero. The verdict is unchanged whether the bar is the ten-run +0.106 or the +0.101 that excludes the degraded-pose run, which is the most that can be said in its favour.

5.4 The bias the replicate was run to test

An unlucky peptide-free pose *inflates* a drop, because the drop is complex minus apo. The top of this ranking is therefore enriched for apo-prediction failures as well as for genuine peptide reading, and the apo term carries 4–10× the complex term's spread, precisely because removing the peptide leaves the binder's placement underdetermined. Replication is the test: artifacts regress toward the middle and real signal does not. The two leaders split on it. `mut_proximal_16` returned +0.112 twice; `mut_proximal_26` fell from +0.109 to +0.081 and is ranked on the pooled mean.

6 The finalists against the wild-type peptide

All twenty submitted designs were co-folded against `VVVGAGGVGK`. Both peptide conditions are written into a *single* invocation for every finalist and for the control, and the whole pass was run twice as independent invocations: 420 predictions, 0 failures. The pairing is the point. The control's bar is formed inside one invocation, so a design's difference has to be formed the same way, or the run offset stays in the difference and out of the error bar.

The control's own difference is $+0.0018 \pm 0.0065$ over the two invocations (+0.0030, +0.0007). It is still positive — the predictor prefers the peptide the receptor does not recognise — but it is tied with zero at this sample size, so the bar is weak and so is any ranking against it. Three of twenty finalists sit below it, thirteen are tied, four are above; seven of twenty are negative in raw sign. Across designs the difference averages $+0.0090 \pm 0.0058$, a spread over designs rather than over runs. The between-run SD is 0.0092 against a within-run 0.0093, leaving a run effect of approximately zero.

The unpaired pass did not reproduce

An earlier pass over the same twenty designs was completed and then withdrawn before it was reported. Its difference took the wild-type mean from one invocation and the G12D mean from others. Measured that way, `bulge_core` — the variant whose interface was placed over peptide positions 5–8, the conformational bulge itself — appeared 4 of 4 negative at a mean of −0.0149, with Fisher exact $p = 0.0072$ against 3 of 16 elsewhere. That was the one result in this campaign that looked mechanistic rather than incidental.

Paired, the same four designs average +0.0115 at $p = 0.1011$, because `bulge_core_27` moves across zero to +0.0767.

Table 6: Grouped by generation variant, paired against the unpaired claim. The grouping was chosen after reading the per-design table and is post-hoc in both columns.

Variant	Hotspots	n	Mean wt − G12D (paired)	Unpaired claim
`mut_proximal`	5, 6	5	−0.0012	−0.0073
`bulge_wide`	4, 5, 6, 7, 8	8	+0.0106	+0.0052
`bulge_core`	5, 6, 7	4	**+0.0115**	**−0.0149**
`hub`	7, 8	3	+0.0183	+0.0209

The measured order is `mut_proximal` < `bulge_wide` < `bulge_core` < `hub`, not the pre-registered one. The only ordinal fact that survives is that `hub`, which puts the least area on the peptide, is the most positive group (0 of 3 negative).

This is not evidence of selectivity, and no design was re-ranked on it. The bar's sign is inverted on the only molecule known to discriminate these peptides; the bar is tied with zero on two invocations; and the grouping is post-hoc. What the replication bought was the retraction of a result that looked mechanistic and was an artifact of its own denominator — the same lesson the peptide-dependence ranking had already learned, applied one level deeper.

7 What is not established

- **Selectivity is sampled, not engineered.** The only chemically distinguishing atom has a 0.013% escape cone; the available discrimination is a sub-Ångström backbone shift below the predictors' coordinate error.
- **Only one co-fold predictor was used.** The standing discipline on this project is that every load-bearing claim gets an independent second method, and that is not met: the Chai-1 container was never located, so the consensus pass did not run. Every number here is Boltz 2.2.1 scoring structures Boltz 2.2.1 predicted. Replication — ten invocations of the bar, two of the cohort — bounds sampling and run-to-run error but cannot detect a systematic error shared by every run, and the tier-1 inversion is exactly such a case.
- **The wild-type pass was never replicated.** One invocation, so the tier-1 column has no between-run denominator — the one thing the drop needed before its own verdicts could be trusted.
- **The aspartate rotamer any design must read has never been observed unliganded.** 7OW4, the receptor-free crystal, has no coordinates for peptide positions 5 and 6; 7OW6 holds the peptide in a conformation a TCR selected. Every design is built against a receptor-shaped conformation.
- **Affinity is not addressed.** The measured quantity is peptide-*dependence*. No K_D is predicted and nothing ranks by binding strength; a design tied with the control on dependence may bind far more weakly.

- **No wet-lab measurement of anything.** No K_D, no expression, no sensorgram.
- **No design was shown to exceed the control on any tier.** That the receptor loses 0.106 ipTM when the peptide is removed establishes the scale a design is judged against. It does not establish that a design matching that number binds the peptide the way the receptor does.

Reproducing the numbers

Every figure in this report is read from a committed file, not retyped: `data/cofold_replicates.csv` (per-invocation rows), `data/cofold_spread_pooled.csv` (the ranking table of record), `data/cofold_wildtype.csv` (the tier-1 column), `submission/pmhc_submission_evidence.csv` (the twenty submitted designs with their evidence) and `docs/experiments/pmhc/verdicts.yaml` (one row per experiment, with its gate and verdict). The commands that regenerate them are in the project README.